

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and executed experiments for reagent systems and assays, analyzing data and presenting findings to technical review boards to support product development.
- Investigated and troubleshooted technical issues within laboratory procedures, contributing to the resolution of complex problems to ensure robust product quality.
- Developed and documented system-level requirements aligned with program goals, ensuring alignment between scientific objectives and product specifications.
- Prepared comprehensive technical reports and study summaries in compliance with **good laboratory practice (GLP)** standards.
- Collaborated with cross-functional teams, including R&D; and quality assurance, to achieve program milestones and meet product launch commitments.
- Mentored junior laboratory personnel on experimental protocols and techniques, guiding them in data analysis and troubleshooting complex applications.
- Integrated customer workflow requirements and emerging scientific advancements into experimental design, ensuring product relevance and market alignment.

SKILLS

Scientific Domain – Immunology, Cell Biology, Clinical Laboratory Science, Reagent Systems, Assays, GLP, Experimental Planning, Data Analysis, Troubleshooting

Programming & Scripting – Python, R, SQL

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, pandas, NumPy, scikit-learn, Statistical Analysis, Hypothesis Testing, A/B Testing

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Enterprise Systems – Java Spring Boot, REST APIs, PostgreSQL, MySQL, Oracle, ERP Systems, Change Management

Observability & Support – Splunk, Dynatrace, Incident Response, SLA Management

PROJECTS

Biomarker Assay Development for Clinical Diagnostics

- Developed and validated a novel ELISA assay for detecting specific biomarkers, achieving a **detection limit of 5 pg/mL** and demonstrating high specificity.
- Designed experimental protocols for assay optimization, including buffer composition and incubation times, using **statistical analysis** to determine optimal parameters.
- Documented all experimental procedures and results in a formal technical report, adhering to **GLP** guidelines for potential future regulatory submission.

Automated Data Analysis Pipeline for High-Throughput Screening

- Engineered a **Python**-based pipeline to automate the analysis of high-throughput screening data, processing ~5,000 data points per run.
- Integrated **pandas** and **NumPy** for data manipulation and applied **scikit-learn** for initial anomaly detection in experimental results.
- Generated interactive **Power BI** dashboards to visualize screening results and trends, enabling faster interpretation by the research team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)